

Problem 1. How many roots does the equation $z^7 - 2z^5 + 6z^3 - z + 1 = 0$ have in the disk $|z| < 1$?

Solution. For $|z| = 1$,

$$|z^7 - 2z^5 - z + 1| \leq 5 < 6 = |6z^3|,$$

so Rouché's theorem implies that $z^7 - 2z^5 + 6z^3 - z + 1 = 0$ has the same number of solutions in the unit disk as $6z^3 = 0$, namely $\boxed{3}$. $\square$

Problem 2. How many roots of the equation $z^4 - 6z + 3 = 0$ have their modulus between 1 and 2?

Solution. Observe that

$$|-6z + 3| \leq 6 \cdot 2 + 3 = 15 < 16 = |z^4|$$

when $|z| = 2$, so the given equation has all four roots in the disk centered at 0 with radius 2. On the other hand,

$$|z^4 + 3| \leq 4 < 6 = |6z|$$

when $|z| = 1$, so the given equation has one root inside (or on the boundary of) the disk centered at 0 with radius 1. Therefore $4 - 1 = \boxed{3}$ of the solutions to the given equation have modulus between 1 and 2. $\square$

Problem 3. How many roots of the equation $z^4 + 8z^3 + 3z^2 + 8z + 3 = 0$ lie in the right half-plane?

Solution. Let $R > 0$ be a real number. The image under the given polynomial f of the line segment joining iR and $-iR$ is contained in the right half-plane because for any $r \in \mathbb{R}$, $f(ir) = (r^4 - 3r^2 + 3) + i(-8r^3 + 8r)$ and $r^4 - 3r^2 + 3 = (r^2 - \frac{3}{2})^2 + \frac{3}{4} > 0$ when r is real, so this image is path homotopic to the segment joining $f(iR)$ and $f(-iR)$. This means that the winding number of the image under f of the loop formed by the segment joining $-iR$ and iR and the counterclockwise semicircular arc from $-iR$ to iR is the same as the winding number of the loop formed by the image under f of the semicircular arc and the segment joining $f(iR)$ to $f(-iR)$.

Parametrize the semicircular arc as a function on $[-\pi/2, \pi/2]$ in the obvious way and parametrize the segment from $f(iR)$ to $f(-iR)$ as a function on $[0, \pi]$; the concatenation of these two paths is a loop which spends half its time traversing the semicircle and half its time traversing the segment. Reparametrize the loop by scaling and shifting its domain to be $[0, 1]$, and call the loop $\gamma: [0, 1] \rightarrow \mathbb{C} - \{0\}$ (to be precise, γ traverses the semicircle on $[0, 1/2]$ and the segment on $[1/2, 1]$); recall that our goal is to compute the winding number of γ around 0. Let $\omega: [0, 1] \rightarrow \mathbb{C} - \{0\}$ be the loop $\omega(t) = R^4 e^{8\pi i t}$ for $0 \leq t \leq 1/2$ and $\omega(t) = R^4$ for $1/2 \leq t \leq 1$, so that ω winds around 0 twice before staying for a while at R^4 . Then,

$$|\gamma(t) - \omega(t)| < |\omega(t)| = R^4$$

for large enough R , noting that $f(\pm iR) - R^4 = (-3R^2 + 3) \pm i(-8R^3 + 8R)$ is only cubic in R so that the inequality holds for $t \in [1/2, 1]$. It follows from a similar argument to Rouché's theorem that γ and ω have the same winding number of $\boxed{2}$.

To be thorough, the inequality tells us that for each t , the segment joining $\gamma(t)$ to $\omega(t)$ does not contain 0, so there is a homotopy $H_s(t): [0, 1] \times [0, 1] \rightarrow \mathbb{C} - \{0\}$ such that for a given t , $H_s(t)$ is the segment from $\gamma(t)$ to $\omega(t)$. This map has the property that for each s , $H_s(0) = H_s(1)$, so H descends to a homotopy $S^1 \times [0, 1] \rightarrow \mathbb{C} - \{0\}$ via the quotient map $[0, 1] \rightarrow S^1$. Such a homotopy preserves winding numbers because the change of basepoint isomorphism from $\pi_1(\mathbb{C} - \{0\}, \gamma(0))$ to $\pi_1(\mathbb{C} - \{0\}, \omega(0))$ along the path $H_s(0): [0, 1] \rightarrow \mathbb{C} - \{0\}$ then sends the homotopy class of γ to the homotopy class of ω . $\square$